Pharmacy Distribution ERP — Administrator Setup & Configuration Guide

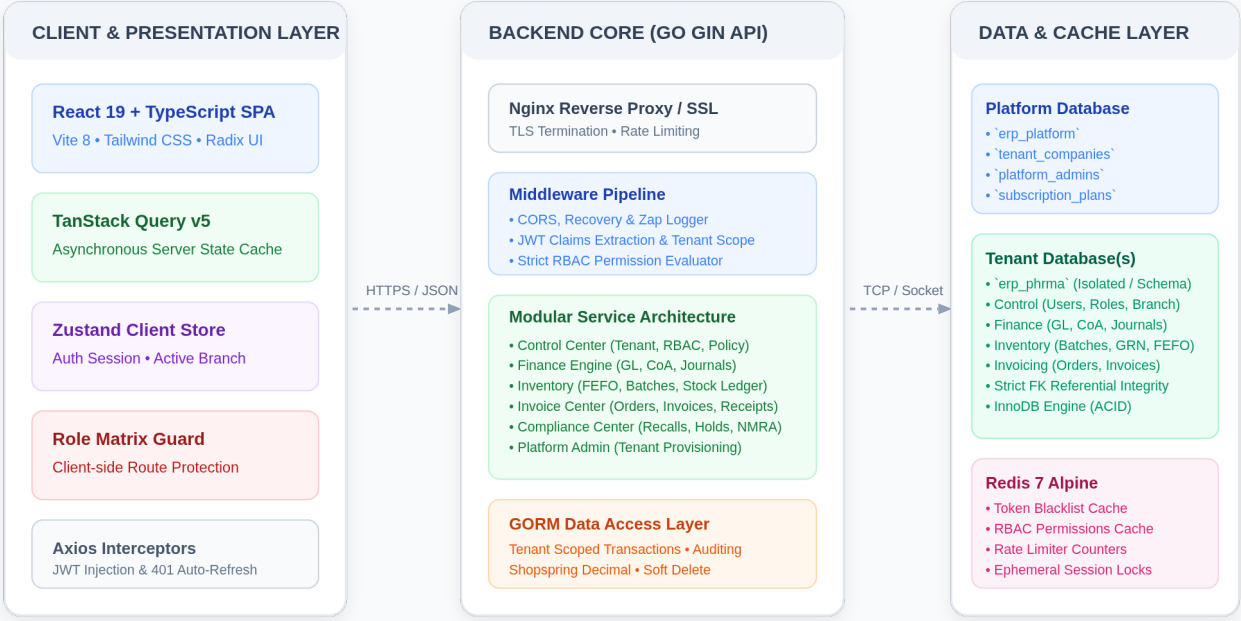
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Target Audience: Platform Administrators, Company Super Administrators, IT Implementation Teams
Version: 1.0.0 | **Publication Date:** 2026-10-15
Verified Platform Environment: Bluehost VPS (`50.6.230.198`), Go 1.23, React 19, MySQL 8.0, Redis 7

1. Introduction & Overview

This guide provides definitive, step-by-step instructions for deploying, provisioning, and administering the **Pharmacy Distribution ERP**. It covers the dual administrative roles: 1. **Platform Administrator:** Responsible for global tenant provisioning, database isolation, subscription plan management, and global infrastructure health. 2. **Company Super Administrator:** Responsible for organizational configuration, branch hierarchy, chart of accounts setup, fiscal year definition, Sri Lankan tax settings, user onboarding, and role-permission matrices.

Figure 1: Pharmacy Distribution ERP — Overall System Architecture

Multi-Tenant High-Availability Enterprise Architecture (React 19 + Go Gin 1.23 + MySQL 8.0 + Redis 7)



2. Infrastructure Prerequisites & Environment Setup

2.1 Hardware and Runtime Requirements

Component	Minimum Specification	Recommended Production Specification
Host Operating System	Ubuntu 22.04 LTS / Debian 12	Ubuntu 24.04 LTS Dedicated VPS
CPU / Processor	4 Cores (x86_64)	8 Cores (AMD EPYC / Intel Xeon)
RAM Memory	8 GB ECC RAM	16 GB - 32 GB ECC RAM
Storage	80 GB SSD	250 GB NVMe Storage (RAID 1/10)
Go Runtime	Go 1.23+	Go 1.25.0
Node.js	Node.js v20 LTS	Node.js v22 LTS
Database Engine	MySQL 8.0 Community	MySQL 8.0 Enterprise / Percona Server
Cache Broker	Redis 7.0 Alpine	Redis 7.2 Alpine with AOF persistence
Reverse Proxy	Nginx 1.24+ with SSL (Certbot)	Nginx 1.26+ with HTTP/2 & Gzip

2.2 Environment Variables Configuration

Backend `.env` (`backend/.env`)

```
# Server Configuration
SERVER_PORT=8080
GIN_MODE=release
LOG_LEVEL=info

# Platform Master Database
PLATFORM_DB_HOST=127.0.0.1
PLATFORM_DB_PORT=3306
PLATFORM_DB_USER=erp_master_user
PLATFORM_DB_PASSWORD=SecureMasterPassword_2026!
PLATFORM_DB_NAME=erp_platform

# Default Tenant Database Connection
TENANT_DB_HOST=127.0.0.1
TENANT_DB_PORT=3306
TENANT_DB_USER=erp_tenant_user
TENANT_DB_PASSWORD=SecureTenantPassword_2026!
TENANT_DB_NAME=erp_phrma

# Redis Cache & Session Broker
REDIS_HOST=127.0.0.1
REDIS_PORT=6379
REDIS_PASSWORD=SecureRedisSecret_2026!
REDIS_DB=0

# Security Secrets (Must be distinct 64-char strings)
PLATFORM_JWT_SECRET=c8f18b348d214e5a96d11e89cf7a9b0c4d2e1f3a5b7c9e0d2f4a6b8c0e2d4f6a
JWT_SECRET=f3a5b7c9e0d2f4a6b8c0e2d4f6ac8f18b348d214e5a96d11e89cf7a9b0c4d2e1
JWT_EXPIRATION_HOURS=24
```

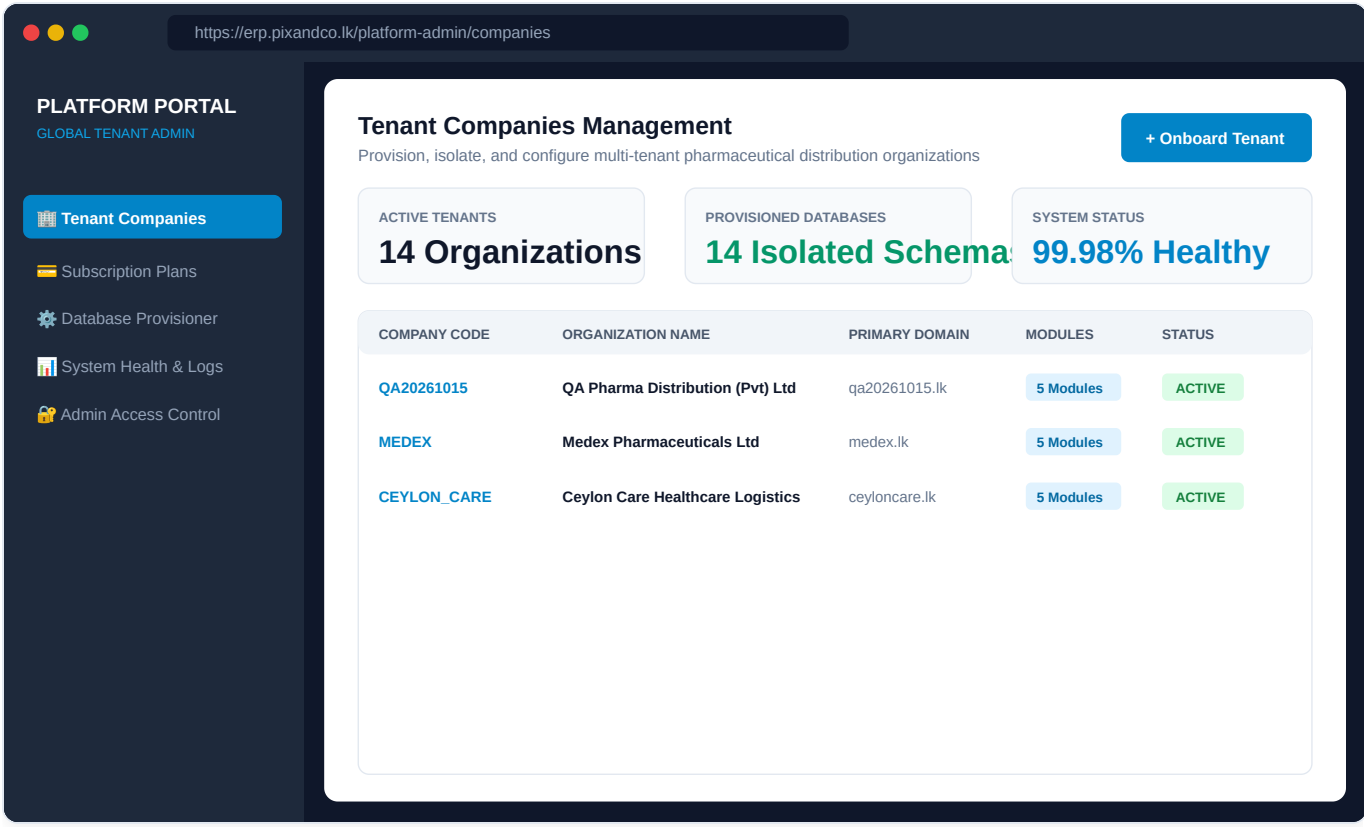
Frontend `.env` (`frontend/.env`)

```
VITE_API_BASE_URL=https://erp.pixandco.lk/api/v1
VITE_APP_NAME="Pharmaceutical Distribution ERP"
VITE_DEFAULT_LOCALE=en-LK
VITE_BASE_CURRENCY=LKR
```

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3. Platform Administration: Onboarding a New Company

Platform administrators operate via the **Platform Portal** at `https://erp.pixandco.lk/platform-admin`.



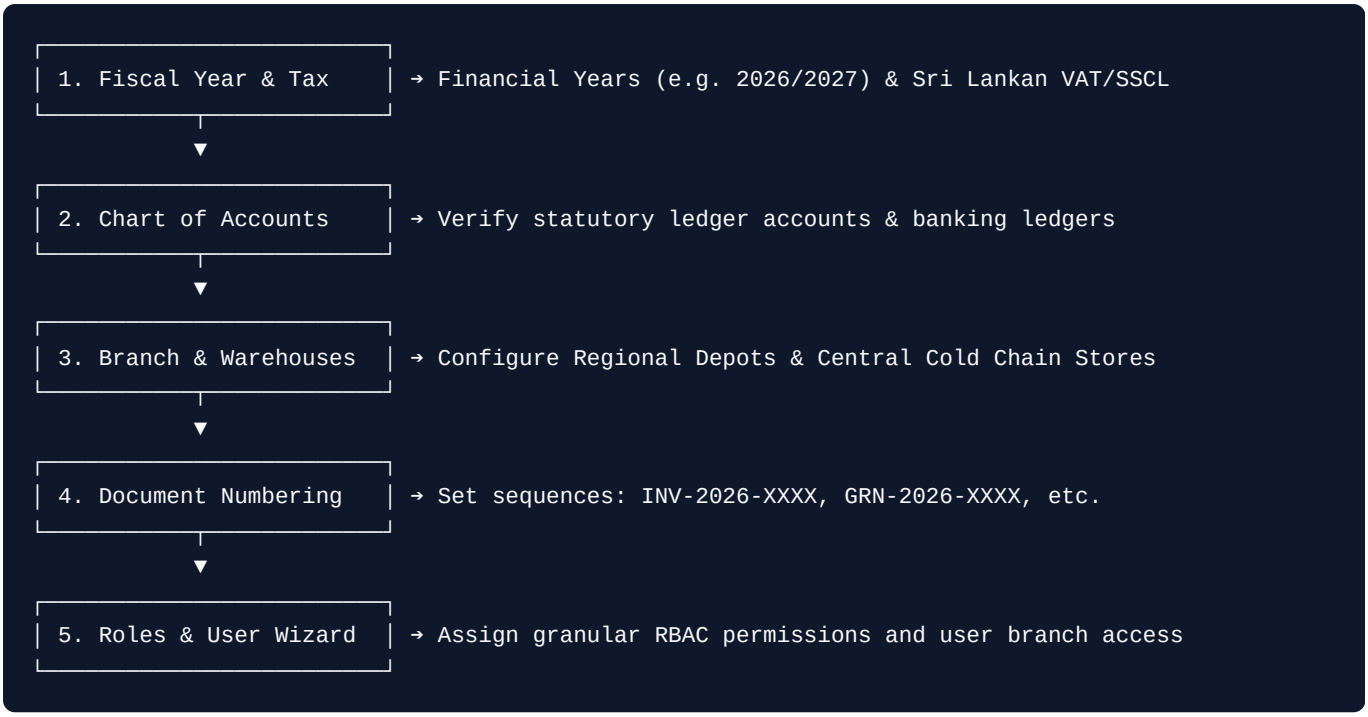
3.1 Step-by-Step Tenant Provisioning

1. Authenticate with platform credentials via `POST /api/v1/platform-admin/login`: `bash curl -X POST https://erp.pixandco.lk/api/v1/platform-admin/login -H "Content-Type: application/json" -d '{ "email": "superadmin@platform.pixandco.lk", "password": "MasterAdminPassword2026!" }'`
2. Invoke Tenant Creation endpoint `POST /api/v1/platform-admin/companies`: `json { "company_code": "QA20261015", "company_name": "QA Pharma Distribution (Pvt) Ltd", "legal_registration_no": "PV-1029384", "contact_email": "admin@qa20261015.lk", "contact_phone": "+94 11 234 5678", "address": "45/2 Galle Road, Colombo 03, Sri Lanka", "base_currency": "LKR", "timezone": "Asia/Colombo", "subscription_plan": "ENTERPRISE", "modules_enabled": [1, 2, 3, 4, 5], "initial_admin": { "full_name": "Dr. Janith Samarasinghe", "email": "admin@qa20261015.lk", "username": "janith_admin", "password": "InitialSecurePassword2026!" } }`
3. **Automated Provisioning Sequence:** The platform service automatically performs: * Record creation in `erp_platform.tenant_companies` with unique company ID (e.g. `2`). * Schema initialization or multi-tenant database binding in `erp_phrma`. * Standard Pharmaceutical Chart of Accounts auto-seeding (25 statutory GL accounts). * Default Headquarters Branch (`MAIN`) and Administration department creation. * Pharmaceutical unit seeders (`Box`, `Bottle`, `Vial`, `Strip`, `Ampoule`). * Super Administrator user creation with salted bcrypt password. * Full mapping in `user_software_access` for Control, Finance, Inventory, Invoice, and Compliance modules.

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4. First-Time Company Setup Sequence

Once a company is provisioned, the **Company Super Administrator** must complete configuration in the following order:

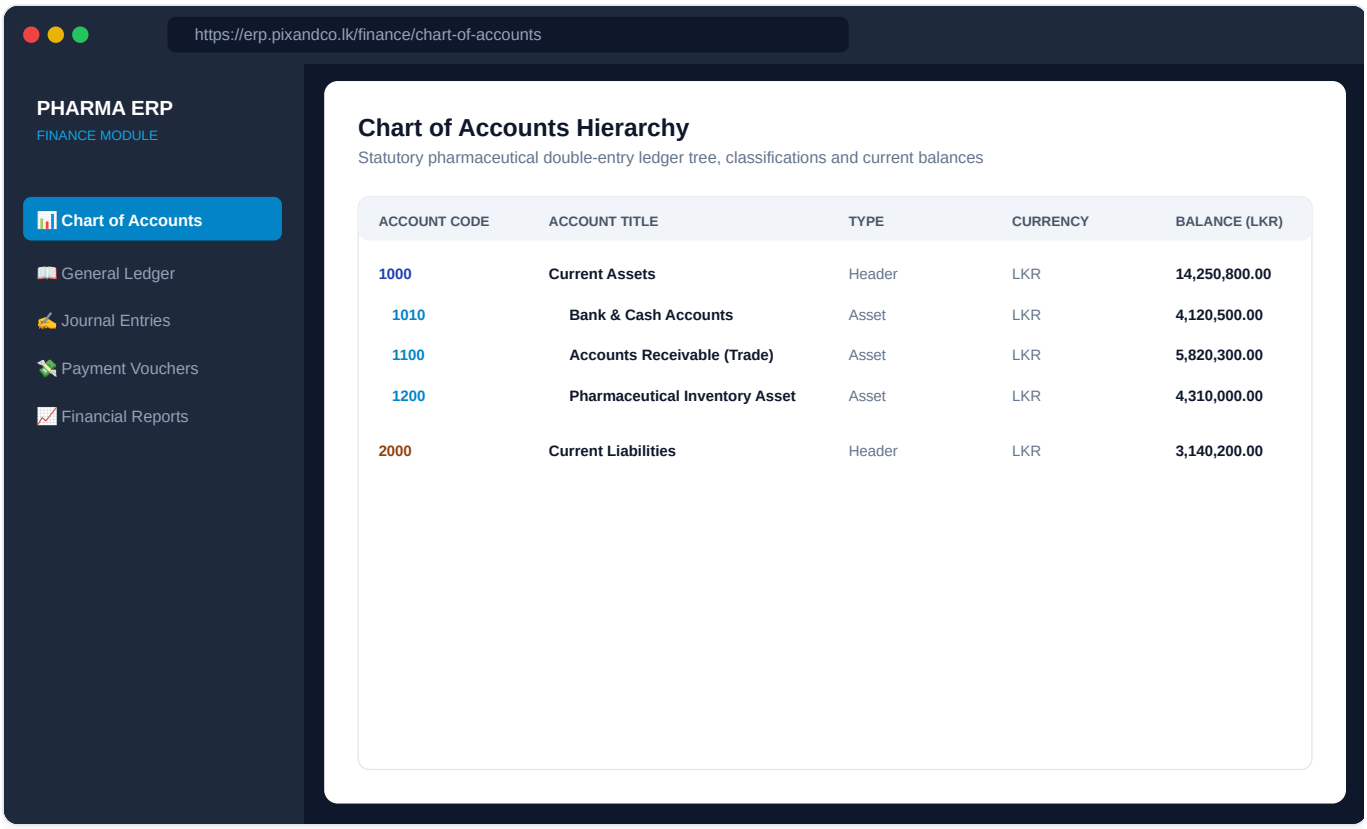


4.1 Step 1: Fiscal Year & Tax Settings

1. Navigate to **Finance ➔ Setup ➔ Financial Years**.
2. Create active financial year: * **Name:** `FY 2026/2027` * **Start Date:** `2026-04-01` * **End Date:** `2027-03-31` * **Status:** `OPEN`
3. Configure Sri Lankan Statutory Taxes under **Finance ➔ Setup ➔ Tax Settings**: * **Standard VAT:** Code `VAT_STD_18` | Rate `18.00%` | Compound: `No` | GL Account: `2200 - VAT Output Payable`. * **Exempt Medicine VAT:** Code `VAT_EXEMPT_PHARMA` | Rate `0.00%` | Compound: `No` | For essential medicines listed under NMRA gazettes. * **SSCL (Social Security Contribution Levy):** Code `SSCL_LIABLE` | Rate `2.50%` | Compound: `No` | GL Account: `2210 - SSCL Output Payable`.

4.2 Step 2: Chart of Accounts Verification

Navigate to **Finance ➔ Chart of Accounts**. Verify that default five root classes exist: * `1000 - Assets` (Current Assets, Accounts Receivable, Inventory, Bank Accounts) * `2000 - Liabilities` (Current Liabilities, Accounts Payable, VAT Payable, SSCL Payable) * `3000 - Equity` (Share Capital, Retained Earnings) * `4000 - Revenue` (Pharmaceutical Sales Revenue, Discounts Allowed) * `5000 - Expenses` (Cost of Goods Sold, Warehouse Logistics, Administrative Expenses)



4.3 Step 3: Multi-Branch & Warehouse Configuration

1. Navigate to **Control Center ➔ Branches**.
2. Set up organizational branches: * `MAIN` — Colombo Head Office & Central Hub (`is_hq = true`). * `QA_CMB` — Colombo Central Distribution Depot. * `QA_KDY` — Kandy Regional Distribution Hub.
3. Navigate to **Inventory ➔ Warehouses**.
4. Configure warehouse locations: * `QA_MAIN` — Central Cold Chain Warehouse (Temperature: 2°C - 8°C). * `QA_SECONDARY` — Ambient Pharmaceutical Transit Store (Temperature: 15°C - 25°C).

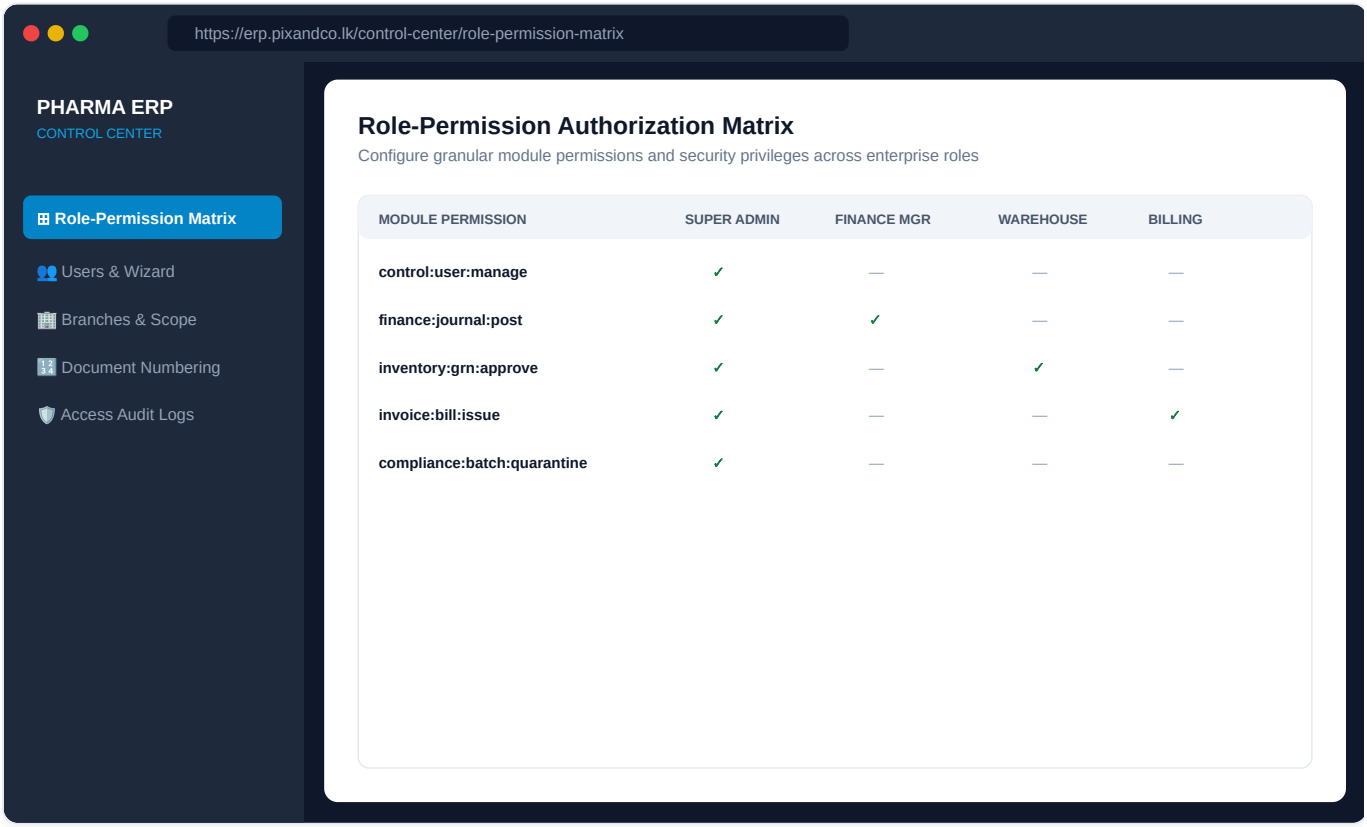
4.4 Step 4: Document Numbering Sequences

Navigate to **Control Center ➔ Document Numbering Rules**. Define standard zero-padded prefixes:

Document Type	Prefix	Current Number	Padding Length	Sample Generated ID
Sales Order	`SO-2026-`	1	4	`SO-2026-0001`
Sales Invoice	`INV-2026-`	1	5	`INV-2026-00001`
Goods Receipt Note	`GRN-2026-`	1	4	`GRN-2026-0001`
Customer Receipt	`RCT-2026-`	1	5	`RCT-2026-00001`
Journal Voucher	`JV-2026-`	1	5	`JV-2026-00001`

5. User Onboarding & RBAC Matrix Configuration

User management is operated via a multi-step modal wizard in **Control Center ➔ Users**.



5.1 User Creation Wizard Steps

1. **Step 1: Identity & Credentials:** Full Name, Username, Corporate Email, Temporary Password, Designation, Department.
2. **Step 2: Branch Scoping:** Select default branch and check all accessible branches (e.g. user permitted to transact only in `QA_CMB`).
3. **Step 3: Role & Permission Assignment:** Assign enterprise roles: * `Super Admin`: Unrestricted system access. * `Finance Manager`: GL, journals, CoA, payments, receipts, taxes. * `Warehouse Officer`: GRN receiving, putaway, stock transfers, batch holds. * `Billing Officer`: Orders, invoices, customer receipts, credit limits. * `Compliance Officer`: Batch recalls, quarantine, regulatory audit logs.
4. **Step 4: Module Access Gates:** Toggle software module licenses (Control, Finance, Inventory, Invoice, Compliance).

6. Security Policies & Audit Monitoring

1. Navigate to **Control Center ➔ Security Settings**.
2. Enforce company password complexity: * Minimum password length: 10 characters. * Require numbers, uppercase letters, and special symbols. * Account lockout threshold: 5 consecutive failed login attempts. * Session inactivity timeout: 30 minutes.

3. Monitor access violations in **Control Center ➔ Access Audit Logs**. Every failed authorization check, privilege escalation attempt, or cross-tenant query generates an immutable audit entry with timestamp, user ID, IP address, and requested action.